

Assignment 4: Temporal-Difference Learning¹

(Cliff Walking) Consider the gridworld shown in Figure 1. This is a standard undiscounted, episodic task, with start and goal states, and the usual actions causing movement up, down, right, and left. Reward is -1 on all transitions except those into the region marked “The Cliff.” Stepping into this region incurs a reward of -100 and sends the agent instantly back to the start.

Write a Python class that simulates the gridworld environment. Then, write another Python class that implements the Sarsa and Q-learning methods with ϵ -greedy action selection, $\epsilon = 0.1$. Last, write a Python class that simulates 500 episodes of cliff walking and draws the variation of sum of rewards with episodes.

S	The Cliff										G

Figure 1: Gridworld for cliff walking.

Requirements

1. Follow PEP 8 – Style Guide for Python Code (<https://peps.python.org/pep-0008/>) for your coding solutions.
2. Submit a PDF report with the provided title page template and leave one blank page for marking feedback. No need to copy and paste your code in the report.
3. Submit a zip package of your Python files (.py files) with the appropriate file structure.

1. This assignment is adapted from R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, Cambridge, MA, 2018.